



Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Interactive Media

Games Development (SE4031)

Assignment 03 - Spell Bound

Runes of the Moonwell

IT22033482 - W D R Pradeepa

1. Game Summary

Runes of the Moonwell is an immersive Virtual Reality experience where the player takes the role of a beginner mage exploring an enchanted courtyard. The player must learn ancient spells, collect magical items, complete a training challenge, and pass the final Moonwell Trial to restore the altar and escape through the exit gate.

The game is played in first-person VR across three connected zones: the Start Courtyard, the Training Zone, and the Moonwell Challenge Area. A VR-friendly HUD displays the current objective, spell status, collection counters, voice recognition feedback, and a magic energy bar.

The player wins by completing all objectives in order: reading the spellbook, collecting moon crystals and a rune key, destroying training orbs with voice spells, entering the challenge area, revealing hidden runes, placing rune stones on pedestals, destroying cursed orbs, and activating the Moonwell altar to open the exit gate.

Spellcasting is voice-only during gameplay. The Windows Speech API KeywordRecognizer listens for the three supported commands while the Windows Desktop .exe runs on a PC connected to a VR headset.

Objective Flow

Step	Task	How to Complete
1	Learn the spells	Interact with the Spellbook in the courtyard
2	Gather moon crystals	Grab and deposit 3 crystals into the Collector Box
3	Find the rune key	Collect 1 Rune Key and deposit it
4	Training practice	Say "star pierce" to destroy 3 Training Orbs
5	Enter challenge	Challenge gate opens after training is complete
6	Reveal runes	Say "glow bloom" to reveal 3 hidden runes
7	Place rune stones	Say "rune rise" three times to place stones on pedestals
8	Clear cursed orbs	Say "star pierce" to destroy 6 Cursed Orbs
9	Complete trial	Moonwell altar activates and exit gate opens

2. Spell List

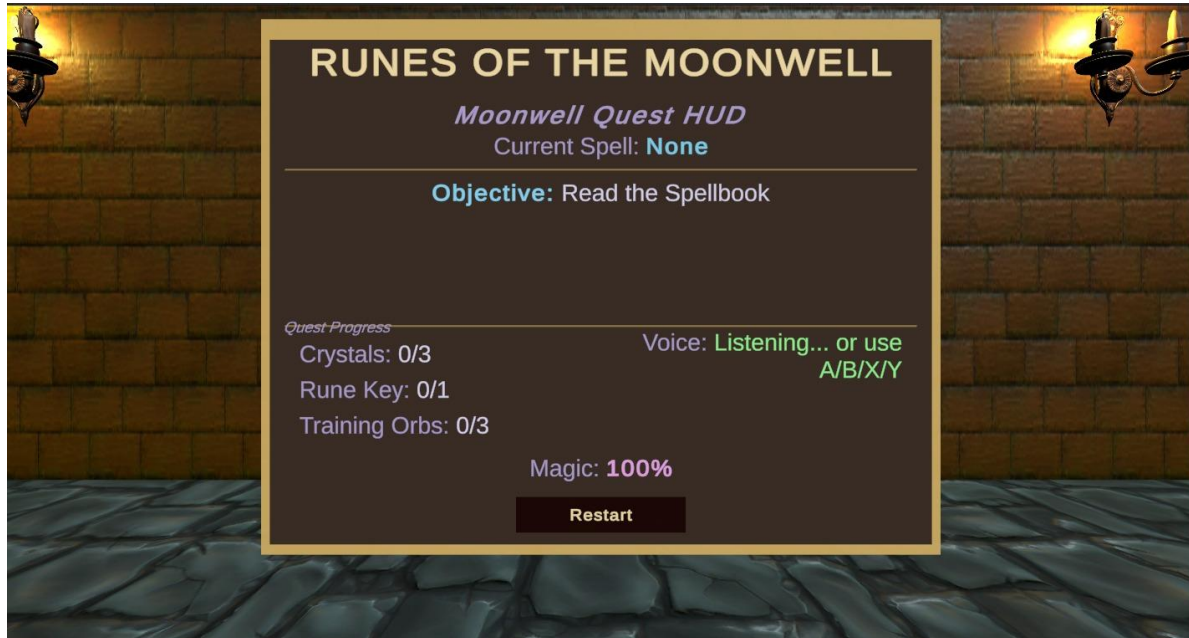
All spells are original. The example spells Fireball, Levitate, and Lumos are not used.

Spell Name	Voice Command	Effect	Visual / Audio Feedback
Star Pierce	"star pierce"	Fires a star-shaped projectile from the wand tip. Destroys Training Orbs and Cursed Orbs on impact.	Projectile VFX, cast sound, orb destruction effect, HUD updates current spell
Rune Rise	"rune rise"	Levitates the nearest unplaced rune stone and moves it onto its assigned pedestal.	Smooth lift animation, cast sound, pedestal placement confirmation
Glow Bloom	"glow bloom"	Activates a magical glow light and particle effect. Reveals hidden runes in the challenge area.	Point light intensifies, particles emit, hidden runes become visible
Open Guide	"open guide"	Opens an in-screen display that contains the instruction of the gameplay.	An in-screen display pops up in the vision, which has written instructions of the gameplay.

Voice System Behavior

- Successful recognition: HUD shows Current Spell: [Spell Name] and Voice: Recognized: [command].
- Unrecognized command: HUD shows Voice: Unknown spell... when the player says an unsupported command such as "fireball".
- Insufficient magic: Spell cannot be cast if the magic bar is empty.
- Magic cost: Each spell costs 5% magic, displayed on the HUD as Magic: XX%.

3. Screenshots of Gameplay





4. Control Guide

VR Headset Controls (PC VR via Link / Air Link)

Action	Control
Teleport movement	Right controller: aim teleport arc at floor, then release to teleport
Smooth walk (optional)	Left controller thumbstick
Grab collectable	Point at item and press Trigger / Grip to grab
Release or drop item	Release Trigger over Collector Box
Interact with spellbook	Point ray at spellbook and press Trigger to open or close lore
Cast spell	Speak the spell command clearly into the PC microphone
Look around	Turn head naturally using VR headset tracking

Voice Commands

Say This	Result
"star pierce"	Fire projectile from wand
"rune rise"	Lift nearest rune stone to pedestal
"glow bloom"	Activate glow and reveal hidden runes
Any other word, e.g., "fireball"	HUD shows unknown spell warning

Testing Without VR Headset (XR Device Simulator)

Action	Keys
Select right controller	Y -> 1
Teleport	Y -> 1 -> hold W -> Space + mouse -> release Left Click on floor
Walk	T -> 1 -> W A S D
Grab object	Point at item with cyan highlight -> hold Left Click
Move hand closer	Hold Space + move mouse

Important Voice Usage Notes

- Voice recognition works in the Unity Editor on Windows and in the Windows .exe build only.
- Voice recognition does not work on standalone VR headset builds such as Quest APK.
- Run the .exe on the PC with the VR headset connected through Link or Air Link.
- Allow microphone access in Windows Settings > Privacy > Microphone.

5. Credits for Assets and Tools Used

Asset / Tool	Source / Usage
Unity Engine	Unity 6 with VR Core Template
XR Interaction Toolkit	Unity Package - VR locomotion, grabbing, and teleportation
OpenXR	VR headset support for PC VR
Windows Speech API	Voice recognition through UnityEngine.Windows.Speech.KeywordRecognizer
TextMeshPro	VR HUD and spellbook lore text rendering
XR Device Simulator	Testing gameplay without a physical VR headset
Audio clips	Pixabay
3D models	Sketchfab, Tripo AI (generated)
Textures / materials	Polligon